Hung-Ting Su

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LLM / MLLM • Multimodal Reasoning • Long-Horizon Evaluation • Reliable AI

Summary

- Researcher-engineer studying multimodal AI safety and how language and multimodal models reason under incomplete, misleading, or adversarial evidence.
- Core interests include trustworthy LLM/MLLM reasoning, long-horizon multimodal evaluation, failure analysis, evidence-seeking behavior, and data-generation or verification workflows.
- Academic research spans trustworthy LLM/MLLM reasoning, long-form video understanding, and multimodal evaluation, developed through work at NTU and Columbia and now extended at NUS to harmful video content detection and adversarial robustness.

Experience

Research Fellow Sep 2026–Present
National University of Singapore Singapore

- Research harmful video content detection and the robustness of multimodal language models to adversarial attacks, including prompt injection, and investigate defenses.

Senior Associate Researcher 2025–Aug 2026
Delta Robotics Innovation Center, Delta Electronics Taiwan

- Built reliable embodied-AI and evaluation infrastructure that connects multimodal data, rollout monitoring, and failure-driven improvement.
- Investigated interfaces between language, perception, action, and emerging world-model directions in VLA-style systems, including how multimodal models support verification.
- Translated research ideas into reproducible experimentation pipelines spanning data collection, analysis, and deployment-oriented debugging.

Postdoctoral Researcher 2023–2025
National Taiwan University (NTU) Taiwan

- Led and mentored projects on trustworthy LLM/VLM reasoning, long-form video understanding, multimodal evaluation, and embodied reasoning under uncertainty.
- Built data-generation, filtering, validation, and benchmark pipelines for long-horizon movie reasoning, false-premise navigation, and planning under underspecified constraints.
- Led projects resulting in CoRL 2024, EMNLP 2024, and ACL 2026, and mentored student-led work accepted at EMNLP 2025, ICCV 2025, ACL 2026, and AAAI 2026.

Visiting Researcher 2022–2023
Columbia University, hosted by Prof. Shih-Fu Chang New York, NY

- Developed zero-shot causal and counterfactual video reasoning frameworks and helped seed later work on trustworthy multimodal evaluation.
- Initiated and coordinated a Columbia–NTU collaboration that led to an EMNLP 2024 publication.

Selected LLM / MLLM Publications

Full publication history: Google Scholar.

- **Unveiling Narrative Reasoning Limits of Large Language Models with Trope in Movie Synopses** *(EMNLP 2024)* [paper] [code] — Analyzes narrative reasoning limits and chain-of-thought hallucinations in LLMs using movie tropes. **Role: Lead**
- **MovieCORE: COgnitive REasoning in Movies** *(EMNLP 2025)* [paper] [project page] — Long-form movie benchmark for cognitive and social reasoning with multimodal models. **Role: Mentor**
- **HERMES: temporal-coHERent long-forM understanding with Episodes and Semantics** *(ICCV 2025)* [paper] [project page] — Improves long-form multimodal understanding through temporal coherence and episodic structure. **Role: Mentor**
- **VLN-NF: Feasibility-Aware Vision-and-Language Navigation with False-Premise Instructions** *(ACL 2026)* [paper] [project page] — Evidence-seeking embodied navigation with explicit abstention under false-premise or infeasible instructions. **Role: Lead**
- **ADAPT: Benchmarking Commonsense Planning under Unspecified Affordance Constraints** *(ACL 2026)* [paper] — Benchmarking and evaluating planning under underspecified affordance and feasibility constraints. **Role: Mentor**
- **Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation** *(CoRL 2024)* [paper] [code] — Zero-shot visual map refinement and semantic replanning for more robust embodied navigation. **Role: Lead**
- **Language Models are Causal Knowledge Extractors for Zero-shot Video Question Answering** *(CVPRW 2023)* [paper] — Zero-shot video QA through causal knowledge extraction with language models. **Role: Lead**

Education

PhD in Computer Science 2023
National Taiwan University (NTU), Advisor: Prof. Winston H. Hsu Taiwan

- Dissertation: *Multi-modal Video Comprehension and Beyond*
- Visiting researcher at Columbia University, hosted by Prof. Shih-Fu Chang.

MS in Computer Science 2017
National Taiwan University Taiwan

BS in Computer Science 2013
National Tsing Hua University Taiwan

Skills

- **LLM / MLLM Reasoning:** chain-of-thought reliability, hallucination analysis, abstention, evidence-seeking reasoning, and tool-using inference.
- **Long-Horizon Multimodal Evaluation:** long-form video understanding, movie reasoning, benchmark construction, data generation, filtering, and automatic verification.
- **Multimodal Systems:** PyTorch, HuggingFace Transformers, VLMs/MLLMs, efficient multimodal models, and multimodal dataset curation workflows.
- **Applied Research Workflow:** Python, Linux, Docker, Git, large-scale experimentation, technical writing, and cross-institution project leadership.

Academic Services

- Area Chair: NAACL/COLING 2027 (ARR 10/2026); ACL 2026 (ARR 01/2026); EACL 2026 (ARR 10/2025).
- Reviewer: ICML, NeurIPS, CVPR, ICCV, NAACL, ACL, EMNLP, CoRL, ICRA, IROS, AAAI, ACM Multimedia, TCSVT.

References

References are available upon request.